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Phase behaviour of essential oil components in supercritical carbon dioxide☆

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Abstract

The vapour–liquid equilibria of α-pinene, limonene and fenchone, and solubility behaviour of camphor were measured in supercritical carbon dioxide (SC-CO_2) as a function of pressure and temperature using the static method. Experiments were carried out in the temperature range of 313–333 K and in the pressure range of 6–13 MPa. Experimental results were correlated by the Peng–Robinson equation of state using conventional mixing rules with one interaction parameter.

Keywords: Camphor; Fenchone; Limonene; α-Pinene; Phase behaviour; Solubility; Supercritical fluids

Nomenclature

AAD	average absolute deviation
N	number of data points
OF	objective function
P	pressure
R	universal gas constant
T	temperature
V	molar volume
x, y	liquid, vapour mole fractions

Greek letters

$\hat{\phi}$	fugacity coefficient
ω	acentric factor

Superscripts

cal	calculated
exp	experimental
liq	liquid phase
s	solid phase
sat	saturation
scf	supercritical phase
vap	vapour phase

Subscripts

2	solute component
b	boiling point
c	critical property
i,j	components i and j

1. Introduction

Separation processes, such as steam distillation and solvent extraction, have been conventionally applied to refine, or to extract essential oils. These processes have several disadvantages, such as the formation of thermally degraded undesirable by-products of essential oils, being energy intensive, leaving solvent residue, and giving lower yields [1–3]. Separation by supercritical fluids,

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having liquid-like densities and good solvent properties, can be considered as an alternative process because of low operating temperatures and leaving no solvent residue in the product. Design and development of supercritical fluid processes depends on the ability to model and predict the solubilities and the phase behaviour of solutes in supercritical solvents. The prediction is usually difficult because of the large differences in structure, size and polarity [4].

Camphor, fenchone, limonene, α-pinene are the major components of the essential oil of lavender flowers (*Lavandula stoechas* subsp. *Cariensis boiss*), and they are widely used in pharmaceutical, food and perfumery applications [5]. Limonene and camphor are also the principal constituents of citrus oil and the essential oil of the camphor tree (*Cinnamomum camphora*), respectively. Many researchers have experimentally investigated the phase behaviour of limonene in CO_2 at high pressures [1,2,6–10]. Some of these researchers have been interested only in the vapour phase, and have not correlated their data with any equation of state (EOS) [6,7]. Tufeu et al. have measured the liquid–gas critical line for a limonene–CO_2 mixture in the temperature range 307.3–349.3 K [11]. The vapour phase solubility values of limonene in CO_2 have been measured by Akgün et al. in the temperature range 308–328 K [12]. The vapour–liquid equilibrium of α-pinene was measured by Pavlicek et al. and Richter et al. [13,14]. However, no data have been reported in the literature on the phase behaviour of fenchone and the solubility of camphor in CO_2 at high pressures.

In this work, the vapour–liquid equilibria (VLE) of α-pinene, limonene and fenchone, and the solubility of camphor in SC-CO_2 were measured at 313–333 K and 6–13 MPa. The results were correlated by the Peng–Robinson equation of state (PR-EOS) using the van der Waals mixing rules with one interaction parameter.

2. Theoretical study

The phase equilibrium equation for a liquid–gas mixture [15] is:

$$\hat{\phi}_i^{\text{liq}} x_i = \hat{\phi}_i^{\text{vap}} y_i, \tag{1}$$

where $\hat{\phi}_i^{\text{liq}}$ and $\hat{\phi}_i^{\text{vap}}$ are the fugacity coefficients for the liquid and vapour phases, respectively.

The solubility, y_2, can be given by the following equation for a gas–solid mixture [4], and it is assumed that CO_2 does not dissolve in the solid phase:

$$y_2 = \frac{P_2^{\text{sat}} \phi_2^{\text{sat}} \exp\left[\int_{P_2^{\text{sat}}}^{P} \left(\frac{V_2^{\text{s}}}{\text{R}T} \right) \text{d}P \right]}{\hat{\phi}_2^{\text{scf}} P}, \tag{2}$$

where subscript 2 indicates the solute component.

The liquid and vapour phase fugacity coefficients, $\hat{\phi}_i^{\text{liq}}$ and $\hat{\phi}_i^{\text{vap}}$, the saturated fugacity coefficient of pure solute, ϕ_2^{sat}, and the fugacity coefficient of solute in the supercritical phase, $\hat{\phi}_2^{\text{scf}}$, are calculated by the Peng–Robinson EOS [16], using van der Waals mixing rules with a single binary adjustable interaction parameter k_{ij} for the mixture to correct for the a_{ij} interaction parameter of the EOS based on the geometric mean mixing rule. k_{ij} was determined by minimizing the objective function (OF) [1] and the average absolute deviation (AAD) [4] defined as:

$$\text{AAD} = \frac{1}{N_x} \sum^{N_x} \left| \frac{x^{\exp} - x^{\text{cal}}}{x^{\exp}} \right| + \frac{1}{N_y} \sum^{N_y} \left| \frac{y^{\exp} - y^{\text{cal}}}{y^{\exp}} \right|, \tag{3}$$

$$\text{OF} = \frac{1}{N_x} \sum^{N_x} \left| x^{\exp} - x^{\text{cal}} \right| + \frac{1}{N_y} \sum^{N_y} \left| y^{\exp} - y^{\text{cal}} \right|, \tag{4}$$

where N is the number of data points.

3. Materials

The test components (supplied by AROMSA LTD, Istanbul) were used without further purification. The purity of test components were more than 99% by gas chromatographic (GC) analysis.

4. Experimental procedure

The vapour–liquid equilibria of α-pinene, limonene and fenchone, and the solubility of camphor

Table 1
Thermophysical properties of pure essential oil components [17,18]

Components	M_W (g mol^{-1})	T_b (K)	T_c [a] (K)	P_c [a] (bar)	ω [b]
α-Pinene (liq.)	136.24	429	634.26	27.24	0.2981
Limonene (liq.)	136.24	449	659.48	27.10	0.3232
Fenchone (liq.)	152.24	466	685.19	29.46	0.3335
Camphor (solid)	152.24	477	701.36	29.46	0.3335

[a] Estimated by the Lydersen group contribution method [19].
[b] Estimated by the Lee–Kesler method [19].

were measured as a function of pressure and temperature in supercritical carbon dioxide (SC-CO_2) using the static method. The physical properties of these test components studied in this work are listed in Table 1. Some of these properties are not available in the literature, so they were estimated by the Lydersen group contribution method and the Lee–Kesler method [19].

The experiments were carried out at temperatures of 313, 323 and 333 K ($\pm$0.1 K) and pressures from 6 to 13 MPa ($\pm$0.01 MPa). Fig. 1 shows the experimental setup which consists of an equilibrium cell (115 ml internal volume) which was placed into a constant temperature heating bath. CO_2 was compressed to the system pressure by a syringe pump (ISCO-model 260D) where pressures were controlled to $\pm$0.01 MPa.

The equilibrium cell was charged with the test components (40 ml) in each run and mixed by a tiny magnetic stirring bar. After the system was stirred for 30 min at the desired operating conditions to reach equilibrium, CO_2 containing solute

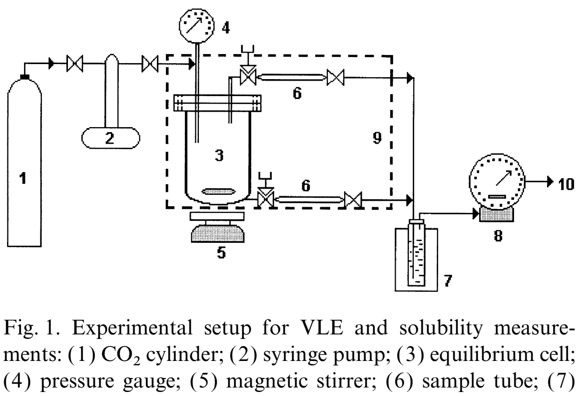

Fig. 1. Experimental setup for VLE and solubility measurements: (1) CO_2 cylinder; (2) syringe pump; (3) equilibrium cell; (4) pressure gauge; (5) magnetic stirrer; (6) sample tube; (7) cold trap; (8) wet-test meter; (9) heating bath; and (10) gas vent.

in the vapour phase and the liquid phase were sent into the sample tubes (2.5 ml internal volume). Before the sample tubes were disconnected from the equilibrium cell, the system was left to reach equilibrium due to a pressure drop of 0.1–0.15 MPa during filling of the sample tubes. While both vapour and liquid samples were separately trapped into the collector containing methanol, the amount of CO_2 consumed was measured by a wet test-meter. After the experiment was terminated, the expansion valves were washed twice with methanol and the concentration of the test-component in each phase was determined by gas chromatographic (GC) analysis.

5. Analytical procedure

The Unicam Model 610 GC was used for the analysis of the test components. The separation was obtained by using a capillary column (EC-WAX Carbowax, 30 m × 0.32 mm i.d., film thickness 0.25 μm). Argon was used as the carrier gas. GC was temperature-programmed as follows: from 40 to 42°C at 0.5°C min^{-1} and then at 5°C min^{-1} to 150°C. The concentration of essential oil components was computed from GC peak areas using calibration curves.

6. Results and discussion

The vapour–liquid equilibrium data were measured for the limonene + CO_2, α-pinene + CO_2 and fenchone + CO_2 systems at 313, 323 and 333 K, respectively. The experimental reproducibility of both vapour and liquid phase mole fractions is

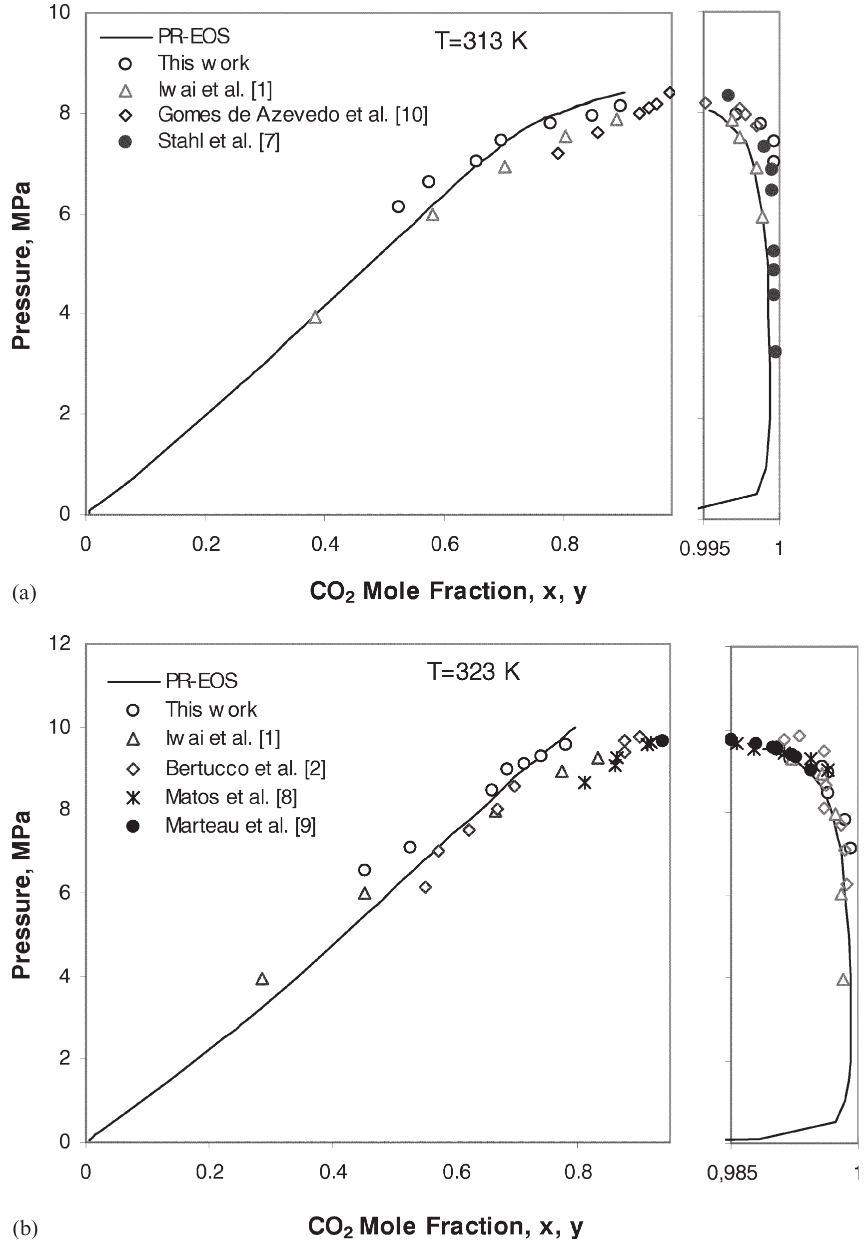

Fig. 2. Vapour–liquid equilibria for CO_2+limonene system: (a) $k_{ij}=0.11$ at 313 K; (b) $k_{ij}=0.11$ at 323 K; and (c) $k_{ij}=0.11$ at 333 K using PR-EOS.

within $\pm 2.8\%$. The experimental data are plotted as a function of pressure, and are illustrated in Figs. 2–4. Fig. 5 illustrates the solubility behaviour of camphor in SC-CO_2. Although, in general, the vapour phase data are in better agreement with the literature than the liquid phase data for the limonene–CO_2 system, there is some deviation between the data of this work and the literature [1,2,6–9]. A plausible explanation could not be found for the deviations. However, in this work,

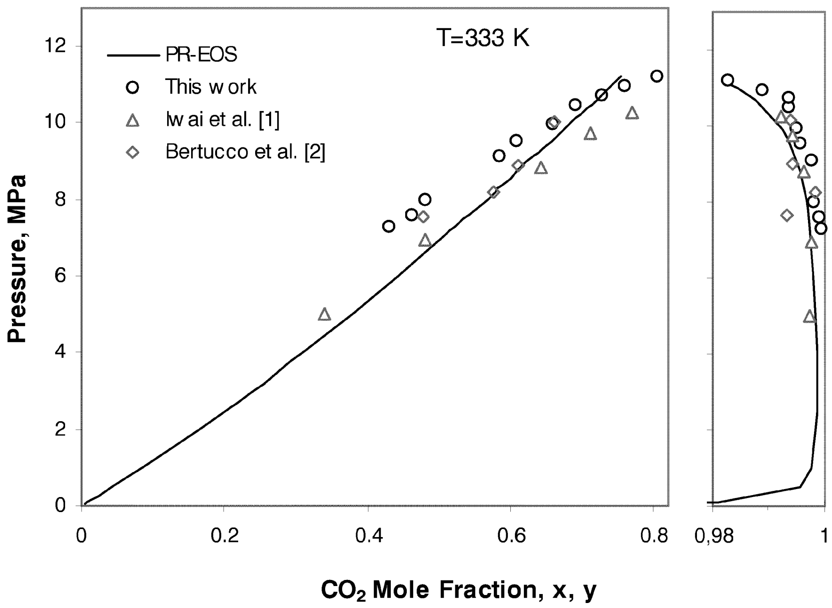

Fig. 2. (*continued*).

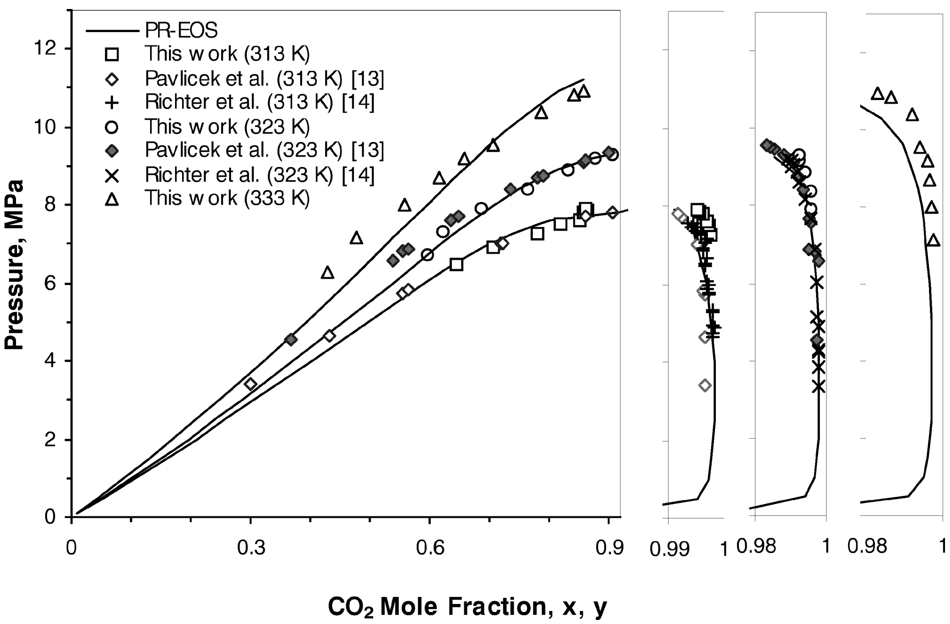

Fig. 3. Vapour–liquid equilibria for $CO_2+\alpha$–pinene system: $k_{ij}=0.11$ at 313 K; $k_{ij}=0.1$ at 323 K; and $k_{ij}=0.11$ at 333 K using PR-EOS.

experiments were repeated several times to get the final data and the reproducibility of phase compositions were within $\pm 2.8\%$, as stated previously.

On the other hand, the liquid and vapour phase compositions of the α-pinene–CO_2 system reported in this work at 313 and 323 K, and the data given

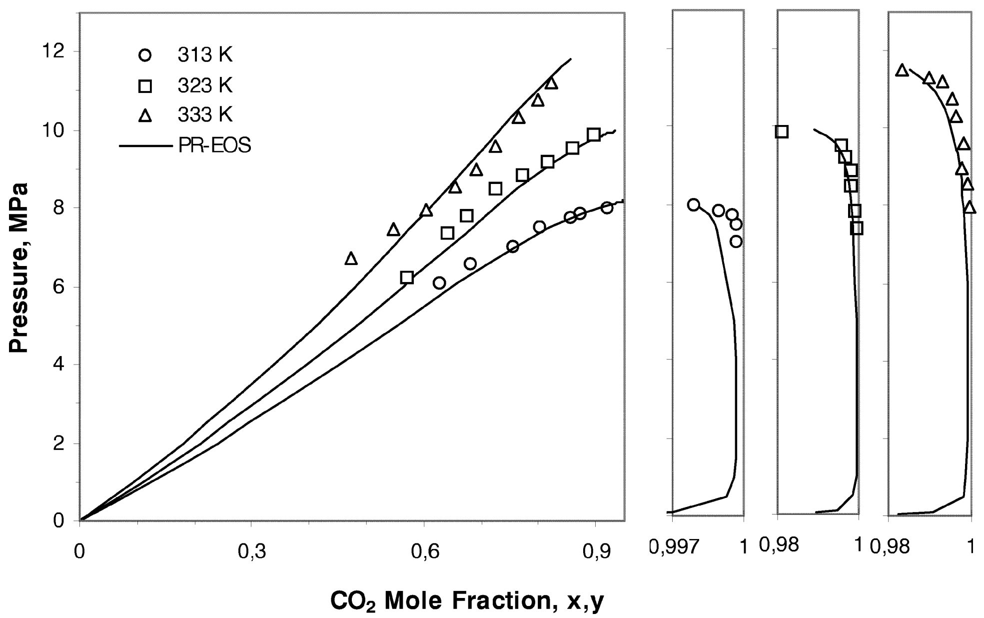

Fig. 4. Vapour–liquid equilibria for CO_2+fenchone system: k_{ij}=0.06 at 313 K; k_{ij}=0.06 at 323 K; and k_{ij}=0.07 at 333 K using PR-EOS.

in the literature [13,14] have almost similar trends. Since no data were reported in the literature for the vapour–liquid equilibria of the fenchone–CO_2 system and the solubility of camphor in supercritical CO_2, a comparison could not be made for these systems. All the data obtained are also reported in Tables 2–5.

The calculated results using the Peng–Robinson EOS are also shown in Figs. 2–5. The binary interaction parameters for solute–CO_2 systems and the statistical data are shown in Table 6. For the solid–gas system, camphor+CO_2, only OF values were used to estimate k_{ij}, and the first summation term on the right-hand side of Eq. (4) was neglected, because the solid phase was assumed to be pure. In general, the correlation of the experimental data of this work by the PR-EOS is acceptable.

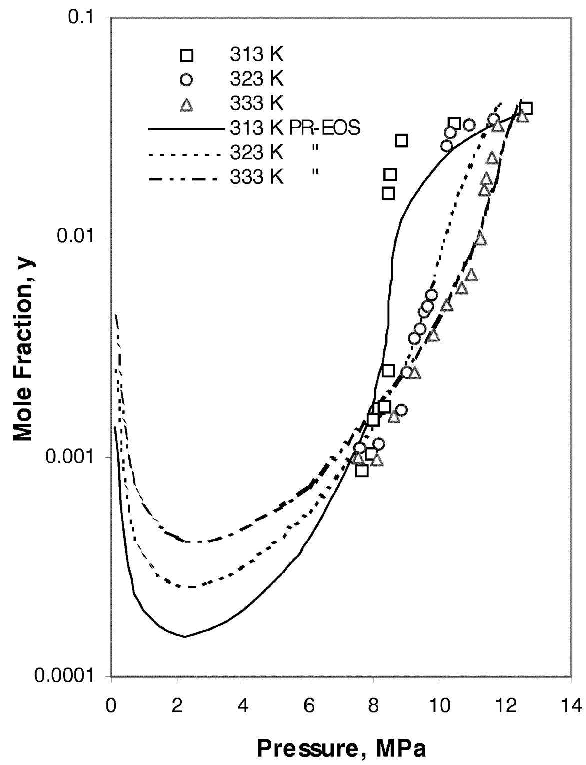

Fig. 5. Solubility of camphor in supercritical CO_2: k_{ij}=0.141 at 313 K; k_{ij}=0.132 at 323 K; and k_{ij}=0.11 at 333 K using PR-EOS.

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Table 2
Vapour–liquid equilibrium data for CO_2 (1)+limonene (2) system

T=313 K			T=323 K			T=333 K		
P (MPa)	x_1	y_1	P (MPa)	x_1	y_1	P (MPa)	x_1	y_1
6.12	0.523	—	6.55	0.455	—	7.29	0.429	0.9994
6.63	0.573	—	7.11	0.528	0.9992	7.60	0.462	0.9991
7.04	0.654	0.9996	7.81	—	0.9984	8.00	0.480	0.9981
7.46	0.693	0.9996	8.47	0.660	0.9965	9.11	0.583	0.9976
7.79	0.778	0.9988	8.97	0.683	0.9964	9.54	0.608	0.9957
7.96	0.847	0.9971	9.12	0.713	0.9957	9.97	0.657	0.9949
8.15	0.894	0.9680	9.31	0.741	0.9919	10.49	0.690	0.9936
			9.56	0.781	0.9905	10.73	0.726	0.9937
						10.95	0.760	0.9890
						11.20	0.804	0.9826

Table 3
Vapour–liquid equilibrium data for CO_2 (1)+α-pinene (2) system

T=313 K			T=323 K			T=333 K		
P (MPa)	x_1	y_1	P (MPa)	x_1	y_1	P (MPa)	x_1	y_1
6.46	0.647	—	6.71	0.597	—	6.29	0.428	—
6.92	0.708	—	7.30	0.622	—	7.15	0.478	0.9974
7.26	0.780	0.9983	7.91	0.686	0.9958	7.99	0.560	0.9970
7.51	0.820	0.9980	8.39	0.765	0.9954	8.69	0.617	0.9963
7.59	0.852	0.9978	8.88	0.833	0.9942	9.19	0.657	0.9960
7.82	0.858	0.9974	9.18	0.878	0.9924	9.54	0.706	0.9938
7.86	0.865	0.9968	9.30	0.908	0.9922	10.40	0.789	0.9919
7.93	0.863	0.9966				10.83	0.844	0.9866
						10.93	0.859	0.9830

Table 4
Vapour–liquid equilibrium data for CO_2 (1)+fenchone (2) system

T=313 K			T=323 K			T=333 K		
P (MPa)	x_1	y_1	P (MPa)	x_1	y_1	P (MPa)	x_1	y_1
6.07	0.626	—	6.23	0.570	—	6.73	0.473	—
6.59	0.682	—	7.38	0.640	0.9996	7.45	0.549	—
7.04	0.756	0.9997	7.83	0.675	0.9992	7.98	0.606	0.9995
7.50	0.802	0.9996	8.49	0.727	0.9983	8.55	0.656	0.9990
7.76	0.858	0.9995	8.87	0.772	0.9981	8.99	0.692	0.9978
7.85	0.873	0.9990	9.20	0.815	0.9966	9.60	0.725	0.9979
8.00	0.920	0.9979	9.53	0.859	0.9956	10.31	0.766	0.9965
			9.87	0.896	0.9808	10.76	0.798	0.9956
						11.20	0.823	0.9932
						11.33	—	0.9897
						11.50	—	0.9831

Table 5
Solubility data of camphor (2) in CO_2 (1) at high pressures

$T=313$ K		$T=323$ K		$T=333$ K	
P (MPa)	$y_2.10^3$	P (MPa)	$y_2.10^3$	P (MPa)	$y_2.10^3$
7.63	0.863	7.55	1.09	7.51	0.990
7.93	1.04	8.14	1.14	8.09	0.970
7.97	1.46	8.82	1.63	8.63	1.52
8.17	1.66	9.00	2.42	9.22	2.43
8.31	1.67	9.22	3.47	9.80	3.60
8.42	2.47	9.43	3.84	10.19	4.97
8.46	15.9	9.54	4.56	10.67	5.87
8.50	19.2	9.66	4.84	10.94	6.73
8.81	27.8	9.77	5.40	11.23	9.93
10.43	32.8	10.12	28.5	11.35	16.4
12.61	38.6	10.20	26.2	11.43	18.7
		10.35	30.1	11.61	23.0
		10.88	32.5	11.78	32.1
		11.66	34.0	12.50	36.0

Table 6
k_{ij} values and statistical data

		313 K	323 K	333 K
α-Pinene	k_{ij}	0.11	0.1	0.11
	AAD	0.046	0.02	0.062
	OF	0.038	0.016	0.04
Limonene	k_{ij}	0.11	0.11	0.11
	AAD	0.099	0.045	0.091
	OF	0.079	0.027	0.051
Fenchone	k_{ij}	0.06	0.06	0.07
	AAD	0.024	0.035	0.035
	OF	0.018	0.026	0.022
Camphor	k_{ij}	0.141	0.132	0.11
	OF	0.005	0.004	0.003

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References

[1] Y. Iwai, T. Morotomi, K. Sakamoto, Y. Koga, Y. Arai, High-pressure vapor–liquid equilibria for carbon dioxide + limonene, J. Chem. Engng Data 41 (1996) 951.
[2] A. Bertucco, G.B. Guarise, A. Zandegiacomo-Rizio, P. Pallado, S. Vieira de Melo, Binary and ternary vapor–liquid equilibrium data for the system CO_2–limonene–linalool, in: E. Reverchon (Ed.), Proc. 4th Italian Cong. on Supercrit. Fluids and their Applications, Capri, Italy, 1997, p. 393.
[3] E. Reverchon, Supercritical fluid extraction and fractionation of essential oils and related products, J. Supercrit. Fluids 10 (1997) 1.
[4] Y.J. Sheng, P.C. Chen, Y.P. Chen, D.S.H. Wong, Calculations of solubilities of aromatic compounds in supercritical carbon dioxide, Ind. Engng Chem. Res. 31 (1992) 697.
[5] N. Adasoglu, S. Dinçer, E. Bolat, Supercritical fluid extraction of essential oil from turkish lavender flowers, J. Supercrit. Fluids 7 (1994) 93.
[6] G. di Giacomo, V. Brandani, G. Del Re, V. Mucciante, Solubility of essential oil components in compressed supercritical carbon dioxide, Fluid Phase Equilibria 52 (1989) 405.
[7] E. Stahl, D. Gerard, Solubility behavior and fractionation of essential oils in dense carbon dioxide, Perfumer Flavorist 10 (1985) 29.
[8] H.A. Matos, E. Gomes de Azevedo, P.C. Simoes, M.T. Carrondo, M. Nunes da Ponte, Phase equilibria of natural flavors and supercritical solvents, Fluid Phase Equilibria 52 (1989) 357.
[9] Ph. Marteau, J. Obriot, R. Tufeu, Experimental determination of vapor–liquid equilibria of CO_2 + limonene and CO_2 + citral, J. Supercrit. Fluids 8 (1995) 20.
[10] E. Gomes de Azevedo, H.A. Matos, M. Numes da Ponte, P.C. Simões, Phase equilibria of systems containing limonene, cineole and supercritical carbon dioxide, in: M. Perrut (Ed.), Proceedings of the First International

Symposium on Supercritical Fluids, Nice, France, 1988, p. 135.
[11] R. Tufeu, P. Subra, C. Plateaux, Contribution to the experimental determination of the phase diagrams of some (carbon dioxide + a terpene) mixtures, J. Chem. Thermodynamics 25 (1993) 1219.
[12] M. Akgün, N. Akgün, S. Dinçer, Solubilities of limonene and camphor in supercritical carbon dioxide, in: E. Reverchon (Ed.), Proc. 4th Italian Cong. on Supercrit. Fluids and their Applications, Capri, Italy, 1997, p. 487.
[13] J. Pavlicek, M. Richter, High pressure vapor–liquid equilibrium in the carbon dioxide–a–pinene system, Fluid Phase Equilibria 90 (1993) 125.
[14] M. Richter, H. Sovová, The solubility of two monoterpenes in supercritical carbon dioxide, Fluid Phase Equilibria 85 (1993) 285.
[15] S.I. Sandler, Chemical and Engineering Thermodynamics, 2nd ed., Wiley, New York, 1989.
[16] D.Y. Peng, D.B. Robinson, A new two constant equation of state, Ind. Engng Chem. Fundam. 15 (1976) 59.
[17] Aldrich Product Catalog, Aldrich Chemical Company, Inc., Milwaukee, WI, 1994, pp. 282, 696, 866, 1147.
[18] R.H. Perry, D. Green, Perry's Chemical Engineers' Handbook, McGraw–Hill, New York, 1985, pp. 3–63.
[19] R.C. Reid, J.M. Prausnitz, T.K. Sherwood, The Properties of Gases and Liquids, McGraw–Hill, New York, 1977.